

Improved Probabilistic Star-Galaxy Separation Method for Rubin LSST Catalogs

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ABSTRACT

The morphological classification of resolved and unresolved sources, commonly referred to as “star-galaxy separation”, is a fundamental step in the analysis of wide-field imaging surveys such as Rubin Observatory’s Legacy Survey of Space and Time (LSST). Star-galaxy separation becomes increasingly challenging at the faint magnitudes probed by LSST because galaxy counts rise rapidly with magnitude while their angular sizes become progressively smaller. LSST photometric catalogs provide a binary classifier **extendedness** in each band, based on the threshold difference between PSF, m_{PSF} , and CModel, m_{CModel} , magnitudes. While this parameter provides a useful baseline for morphological classification, its binary nature prevents the optimal combination of information from multiple bands, as well as from other classification methods (e.g., those based on colors or external datasets). In this work, we implement the Bayesian star-galaxy classification method of Slater, Ivezić & Lupton (2020), which overcomes these shortcomings. In each band, we model the distribution of sources in the two-dimensional space defined by $\Delta m = m_{\text{PSF}} - m_{\text{CModel}}$ and magnitude m_{CModel} . At a fixed magnitude, the distribution is modeled as a mixture of unresolved and resolved components which, with appropriately chosen priors for the star-to-galaxy number ratio, allows us to compute p_S , the posterior probability that a source is unresolved. We apply this method to the Rubin Data Preview 2 object catalog and validate the resulting classifications using external HST labels in the COSMOS field. Using these labels as the ground truth, we quantitatively compare the performance of the p_S -based classifier with that of the Rubin **extendedness** parameter. We find that the p_S -based classifier consistently outperforms the Rubin **extendedness** baseline, with the improvement increasing as additional bands are incorporated. Approximately, the p_S classifier achieves the same classification performance at a magnitude limit about half a magnitude fainter than the **extendedness** classifier. Depending on the scientific application, this morphological classification can be further improved by incorporating complementary color-based classifiers.

Keywords: Galaxies (573) — Cosmology (343) — Stellar astronomy (1583) — Solar physics (1476)

1. INTRODUCTION

The morphological classification of resolved and unresolved sources⁴, commonly referred to as “star-galaxy separation”, is a fundamental step in the analysis of wide-field imaging surveys such as the Rubin’s Legacy Survey of Space and Time (LSST). At bright magnitudes this problem is relatively simple, since stars outnumber galaxies and galaxies that exist are obviously extended on the sky. Star-galaxy separation becomes

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⁴ Note that quasars and galaxies can be unresolved and even some stars can be resolved, such as Asymptotic Giant Branch stars.

increasingly harder at faint magnitudes, such as those probed by Rubin’s Legacy Survey of Space and Time (LSST; for details see [Ž. Ivezić et al. 2019a](#) and references therein) because galaxy counts rapidly increase with magnitude, while their angular sizes become smaller. The selection of clean faint stellar samples is especially hard.

A variety of methods have been proposed in the literature for morphological classification of sources in images. [C. T. Slater et al. \(2020\)](#) (hereafter SIL2020) discussed the statistical foundations of morphological star–galaxy separation and showed that many of the star–galaxy separation metrics in common use today, e.g., by Sloan Digital Sky Survey (SDSS) *photo* pipeline ([R. H. Lupton et al. 2002](#)), or SExtractor pipeline ([E. Bertin & S. Arnouts 1996](#)), are closely related both to each other, and to the Bayesian model odds ratio, first derived by [W. L. Seaborg \(1979\)](#). SIL2020 quantified the scaling of various algorithms with the source noise properties and showed that the Bayesian model odds ratio achieves the best performance. They also showed how to probabilistically combine multiple measurements, either of the same type (e.g., subsequent exposures), or differing types (e.g., multiple bandpasses), or differing methodologies (e.g., morphological and color-based classification).

Following SDSS, LSST photometric catalogs use a binary classifier `extendedness` in each band, based on the threshold difference between PSF, m_{PSF} , and CModel, m_{CModel} , magnitudes, $\Delta m = m_{\text{PSF}} - m_{\text{CModel}}$. For unresolved sources, this Δm is vanishing within noise, with `extendedness`=0, while for resolved objects, selected by $\Delta m > 0.016$ mag, `extendedness`=1. Resolved objects have positive Δm because their PSF flux is smaller than CModel flux due to a narrow weighting PSF profile. For an illustration of Δm distribution in Rubin’s Data Preview 1 catalogs, see Fig. 28 in [Vera C. Rubin Observatory Team et al. \(2026\)](#) or Fig. 3 further below).

Although `extendedness` was successfully used in thousands of papers based on SDSS catalogs, this approach is not optimal for LSST. As discussed in SIL2020, when the star–galaxy separation threshold is optimized at the faint, low S/N, end of the data, the ability to recognize barely resolved objects at the bright end is not fully exploited. Furthermore, the binary nature of `extendedness` parameter prevents optimal combination of information from different bands, as well as from other classification methods (e.g., based on colors, or other datasets). SIL2020 showed how to turn Δm into a probability distribution, which then allows proper probabilistic combination of multiple bands. With appropriately chosen priors for star-to-galaxy counts ratio, the posterior probability that a source is unresolved (or resolved), can be computed in Bayesian framework.

The main goal of this paper is to implement the steps towards Bayesian star–galaxy separation outlined in SIL2020 using Rubin’s Data Preview 2 data. We describe our dataset in §2, and in §3 discuss star–galaxy separation methodology, including quantification of its performance using HST-based labels. We discuss our results and how to retrieve our catalog with improved star–galaxy separation parameters in §4.

2. DATASETS

Here we describe two principal datasets used in our analysis: Rubin Data Preview 2 catalogs and HST data with star–galaxy labels used for testing methods discussed in §3.

2.1. Rubin Data Preview 2

All data collected with the LSST Camera [Vera C. Rubin Observatory Team \(2026\)](#) during the commissioning of the Rubin Observatory will be released as Data Preview 2⁵ (hereafter DP2). Coadded deep images and resulting catalogs have been released in July 2026 as Early Data Preview 2. These coadded images are based on about 24,000 single-visit images and cover about 3,000 sq. deg. of sky.

In our analysis we focus on the COSMOS field because it is both already well observed by Rubin (to only about 1 magnitude shallower limit than anticipated for 10-year LSST), and space-based HST COSMOS2020/FARMER star and galaxy labels are available ([J. R. Weaver et al. 2022](#)). As a control sample for testing code robustness, we also used Rubin data for the Extended Chandra Deep Field South (ECDFS). The Rubin $5 - \sigma$ depth for point sources in the COSMOS field, estimated as the median magnitude of point sources with photometric errors of ~ 0.2 mag, is (25.66, 26.66, 26.36, 25.91, 24.94, 23.59) in the *ugrizy* bands, respectively.

⁵ See <https://dp2.lsst.io/>

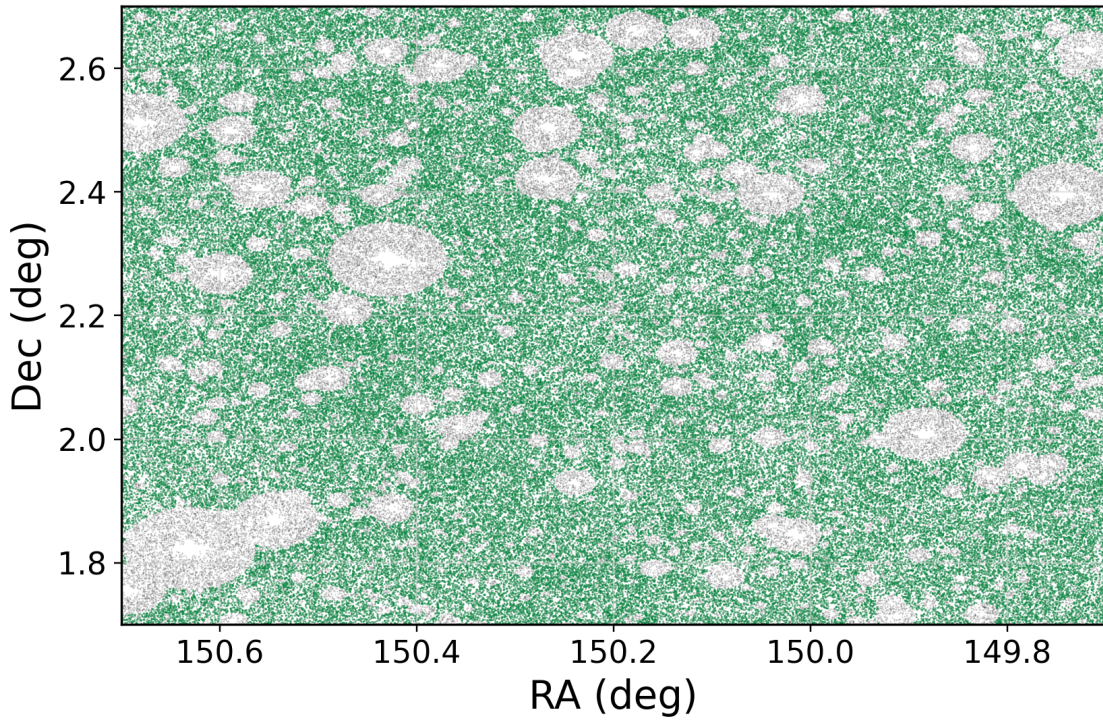


Figure 1. The footprint for the DP2-COSMOS sample analyzed here. Gray points show Rubin DP2 objects and green points show the subset matched to the HST COSMOS2020/FARMER sample a 0.5 arcsec matching radius. The white regions in the centers of gray areas correspond to areas where no objects are listed in either catalog.

2.2. HST Star-galaxy Classification Labels in the COSMOS Field

For the COSMOS validation sample, we select Rubin DP2 sources in the rectangular region $149.7 < \text{RA} < 150.7$ and $1.7 < \text{Dec} < 2.7$, with $16 \leq r_{\text{CModel}} < 26$. Objects listed in the COSMOS2020/FARMER reference catalog are matched to DP2 sources using a 0.5 arcsec matching radius. Within the selected region and magnitude ranges, the DP2 sample contains 269,606 objects, of which 193,586 are matched to the COSMOS2020/FARMER objects and 76,020 are unmatched (a matching fraction of 72%). This somewhat low matching fraction is due to large areas around bright stars that are masked in the COSMOS2020/FARMER catalog (see Figure 1).

The matched validation sample contains 8,788 stars and 184,798 galaxies, classified using HST’s `truth_binary = 1` for stars and `truth_binary = 0` for galaxies. Their number counts as function of magnitude are shown in Figure 2. The low fraction stars is due to very faint sample limit $r \sim 26$) and high Galactic latitude ($b \sim 50^\circ$).

2.3. The color-magnitude distributions of sources classified using HST labels

The external COSMOS2020/FARMER labels provide the reference classification, considered here as ground truth, for examining the distribution of stars and galaxies in various diagrams and for measuring the performance of star-galaxy separation methods. Nevertheless, it is likely that some very faint sources classified by HST as stars might be unresolved galaxies even at HST’s resolution. Therefore, the HST labels will help us assess whether a source should be resolved or unresolved in Rubin images, but do not necessarily provide definitive information about astrophysical nature of a source (e.g., to separate unresolved stars and galaxies, one would need spectra or broad-band photometry over a much wider wavelength range, say, in near-infrared K band). Furthermore, no classification is perfect and thus we first examine the distribution of HST-classified “stars” and “galaxies” in various informative color-color diagrams constructed with Rubin photometry.

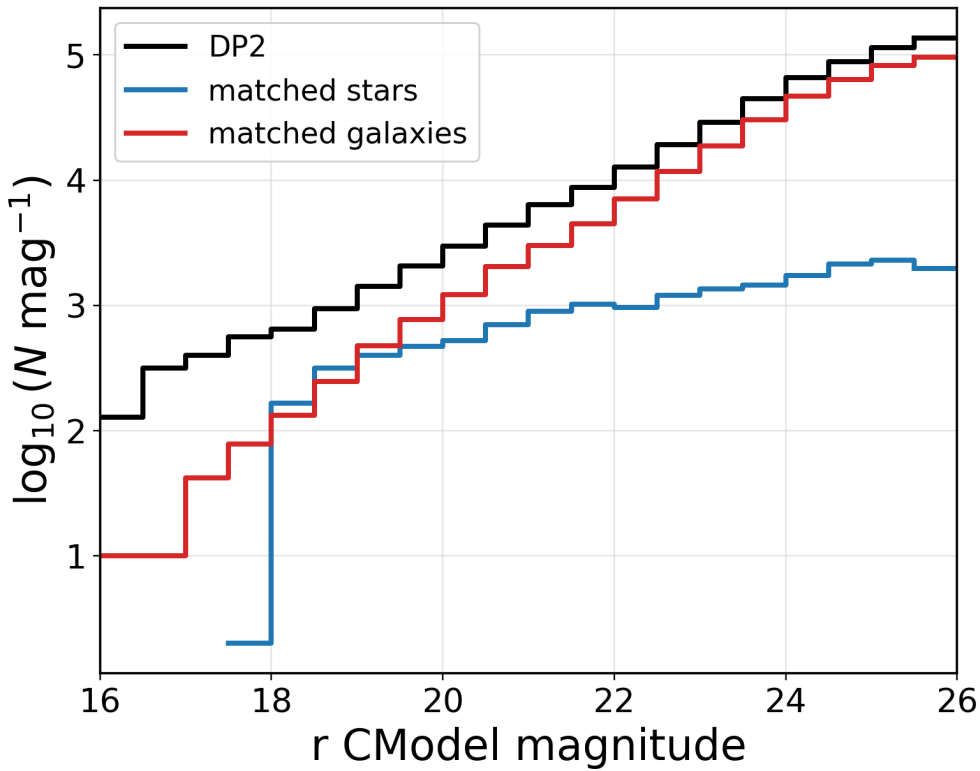


Figure 2. Magnitude distribution of the DP2-COSMOS sample. The black curve shows all selected DP2 sources, while the blue and red curves show matched COSMOS2020/FARMER sources classified as stars and galaxies, respectively. Counts are shown per magnitude, using 0.5 mag wide bins. Note that galaxies outnumber stars at the faint end by more than an order of magnitude.

The top left panel in Figure 3 shows the distribution of stars and galaxies classified using HST labels in the $\Delta m = m_{\text{PSF}} - m_{\text{CModel}}$ vs. m_{CModel} diagram. As we describe in detail in the next Section, modeling this distribution is a key ingredient of the Bayesian star-galaxy separation method presented here.

The top right panel in Figure 3 shows that faint galaxies ($r > 24$), in addition to outnumbering stars at faint magnitudes, have on average bluer $g - i$ colors than stars. The clearly different distributions of stars and galaxies in color-color diagrams shown in the bottom two panels in Figure 3 demonstrate that colors can be used to enhance morphological star-galaxy separation. Nevertheless, Figure 4 and the top left panel in Figure 3 clearly show that increased photometric noise close to the faint magnitude limit will cause deteriorated performance for both morphological and color-based star-galaxy separation methods. Extending the start of this deterioration to a fainter magnitude limit is one of the primary goals of this work. It is notable that the top right panel in Figure 4 implies that most unresolved faint blue sources have $u - g$ color more typical of low-redshift quasars than of main sequence stars (for a detailed discussion of the blue star counts see L. Palaversa et al. 2026).

2.4. The color-magnitude distributions of sources classified using *extendedness* parameter

Difficulties with star-galaxy separation encountered at the faint end are illustrated in Figure 5, which compares subsamples classified using HST ground-truth labels and the default Rubin *extendedness* binary class in the r band. As evident, a significant fraction of faint sources within about two magnitudes from the faint limit are misclassified when using default *extendedness* parameter. Misclassifications have a more severe impact on stellar sample: only 26% of “stars” selected by *extendedness*=1 condition are unresolved in HST data. This high level of stellar sample contamination by faint galaxies increases towards the faint end (see the bottom left panel in Figure 5). As the same time, the selection completeness for “stars” (objects unresolved by HST) is only 67%. Due to much higher counts at faint magnitudes and high Galactic latitudes,

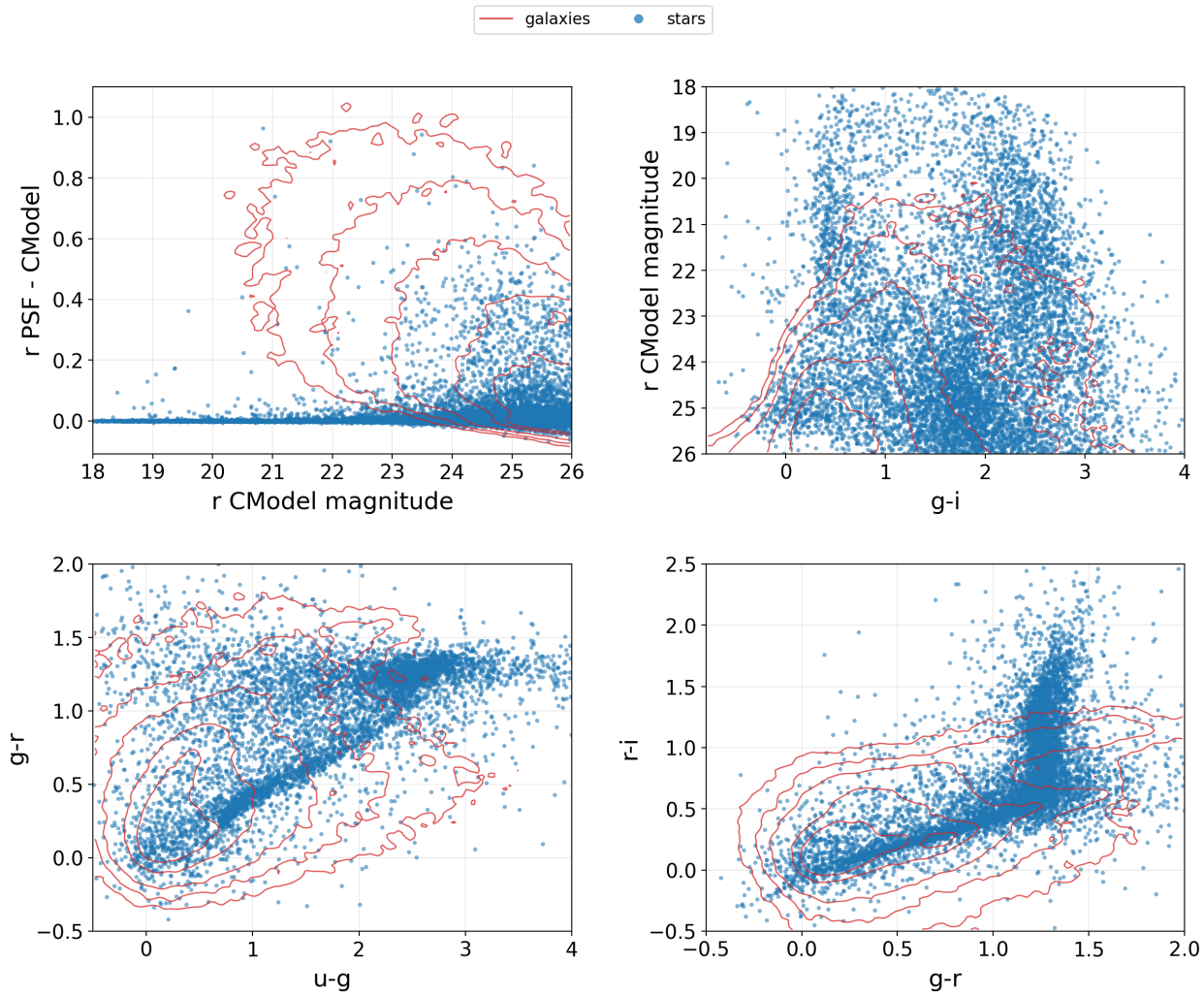


Figure 3. The distribution of stars (blue dots) and galaxies (red contours), classified using HST labels, in various informative diagrams. The CModel-magnitude based colors are corrected for the ISM dust extinction, while the r -band CModel magnitude is not. The top left panel shows the $\Delta m = m_{\text{PSF}} - m_{\text{CModel}}$ vs. m_{CModel} diagram, where stars form a narrow sequence centered on $\Delta m \approx 0$ and for galaxies $\Delta m > 0$. The top right panel shows the r vs. $g - i$ color magnitude diagram and two bottom panels show most informative color-color diagrams. The main stellar locus, dominated by main sequence stars and red giants, is clearly visible with its blue end redwards from $u - g \sim 0.6$; unresolved sources with $u - g < 0.6$ are expected to be a mixture of low-redshift quasars, white dwarfs and unresolved binaries.

the purity and completeness for galaxies are much higher: 98% and 89%, respectively. In other words, a simple classifier that declares every object to be a galaxy would formally perform quite well. Therefore, we will focus our analysis in the next Section on the classifier’s performance when selecting unresolved objects.

2.5. Suspect HST Classification Labels

There is a subset of objects classified as “stars” (unresolved) by HST which appear clearly resolved when using Rubin’s default `extendedness` parameter. Figure 6 shows their distribution in color-color diagrams, with symbols color-coded by Δm in the r band. As evident, their color-color distribution is inconsistent with main stellar locus and similar to that expected for barely resolved or unresolved binary stars (see Figures 2 and 4 in V. Smolčić et al. 2004.) To test this hypothesis, we visually inspected several hundred images (443 objects with HST label “star”, $r < 23$ and $\Delta m > 0.05$ mag; see Figure 7 for an example). Only 1.4% of selected objects are detected as “isolated” (computed using processing flag “detect_isolated”) and about

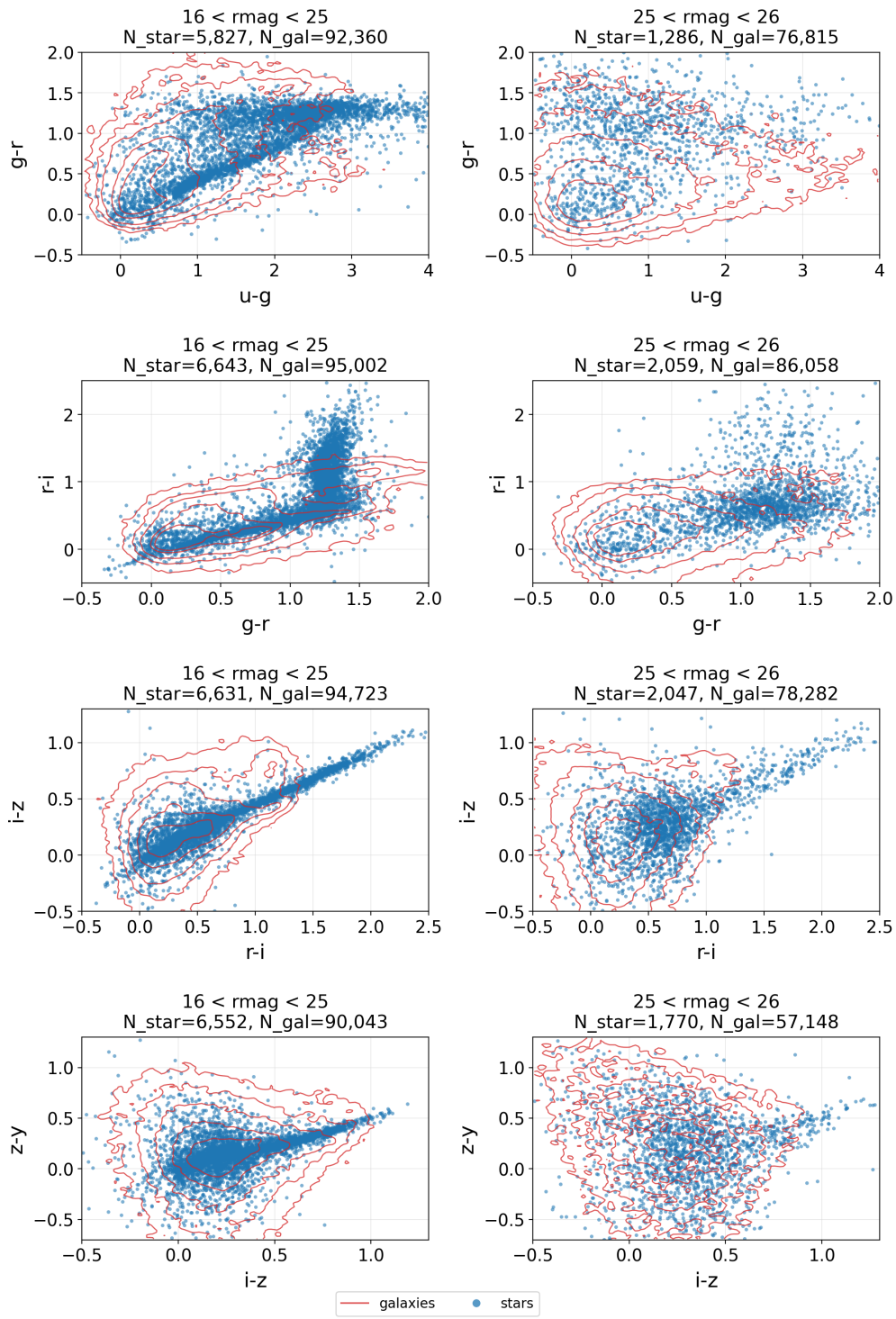


Figure 4. Similar to the bottom two panels in Figure 3, except that here all principal color-color diagrams are shown and the data are split by the r -band CModel magnitude to show deterioration of photometry close to the faint magnitude limit (left: $16 < r_{\text{mag}} < 25$; right: $25 < r_{\text{mag}} < 26$). The $5 - \sigma$ depth for point sources in the r band for this dataset is 26.36.

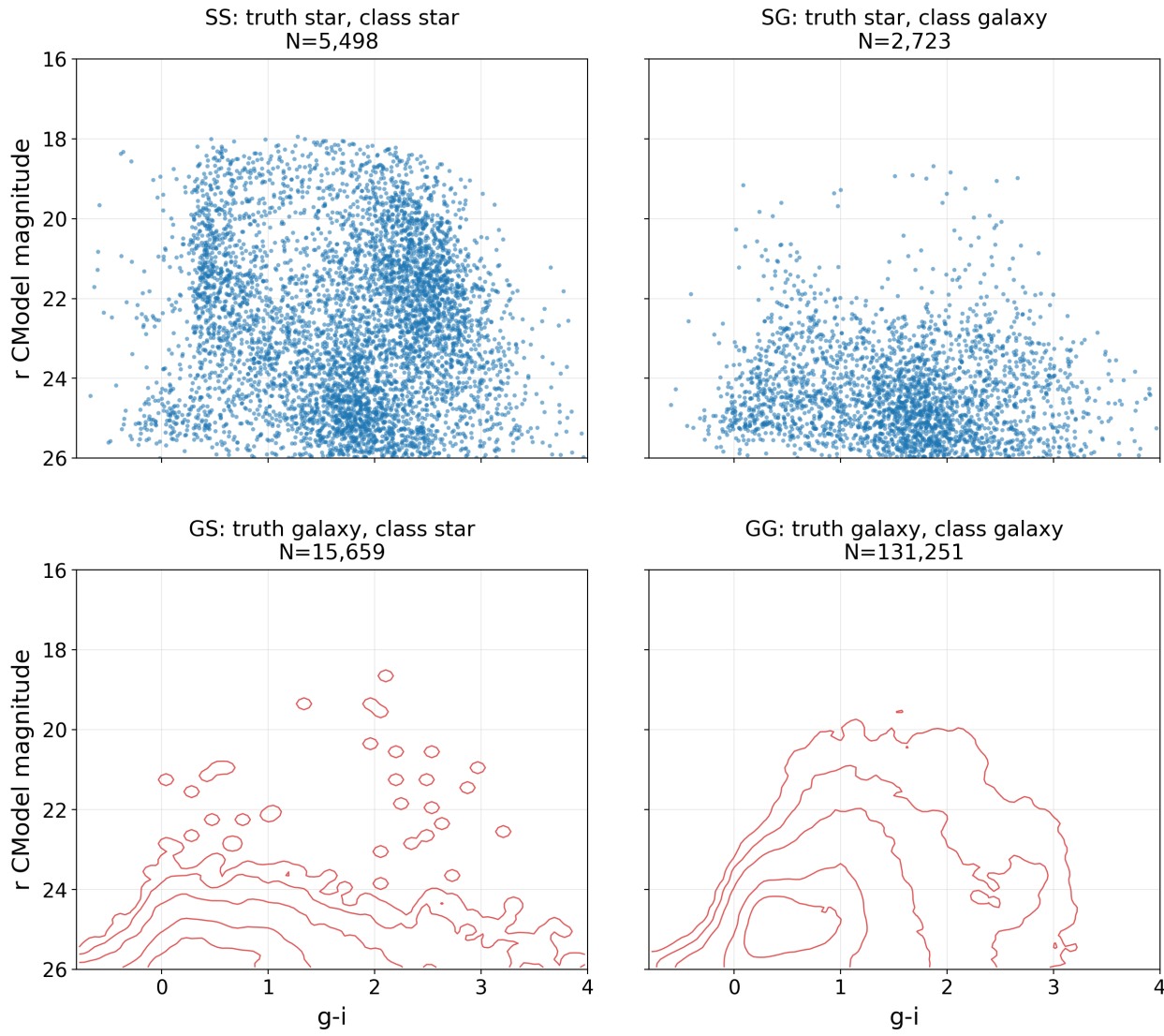


Figure 5. Performance of Rubin’s default **extendedness** parameter in the r band relative to HST ground-truth labels. The top left and bottom right panels show correctly classified unresolved and resolved sources in the r vs. $g - i$ color-magnitude diagram, with counts shown in the title above each panel. The bottom left and the top right panels show misclassified sources. Note that the stellar sample selected by **extendedness**=0 is severely contaminated by galaxies for $r_{\text{CModel}} > 24$ (bottom left panel).

half seem to have a nearby stellar-like companion. In some cases a bright source or an extended galaxy is nearby, and about one tenth or fewer appear bluish and consistent with quasars. Detailed examination of such cases is beyond the scope of this work.

Because we aim to test statistical performance of our star-galaxy separation method on unambiguous objects, we exclude objects with HST “star” label that both appear resolved in Rubin data and also have Rubin colors inconsistent with the main stellar locus.

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APPENDIX

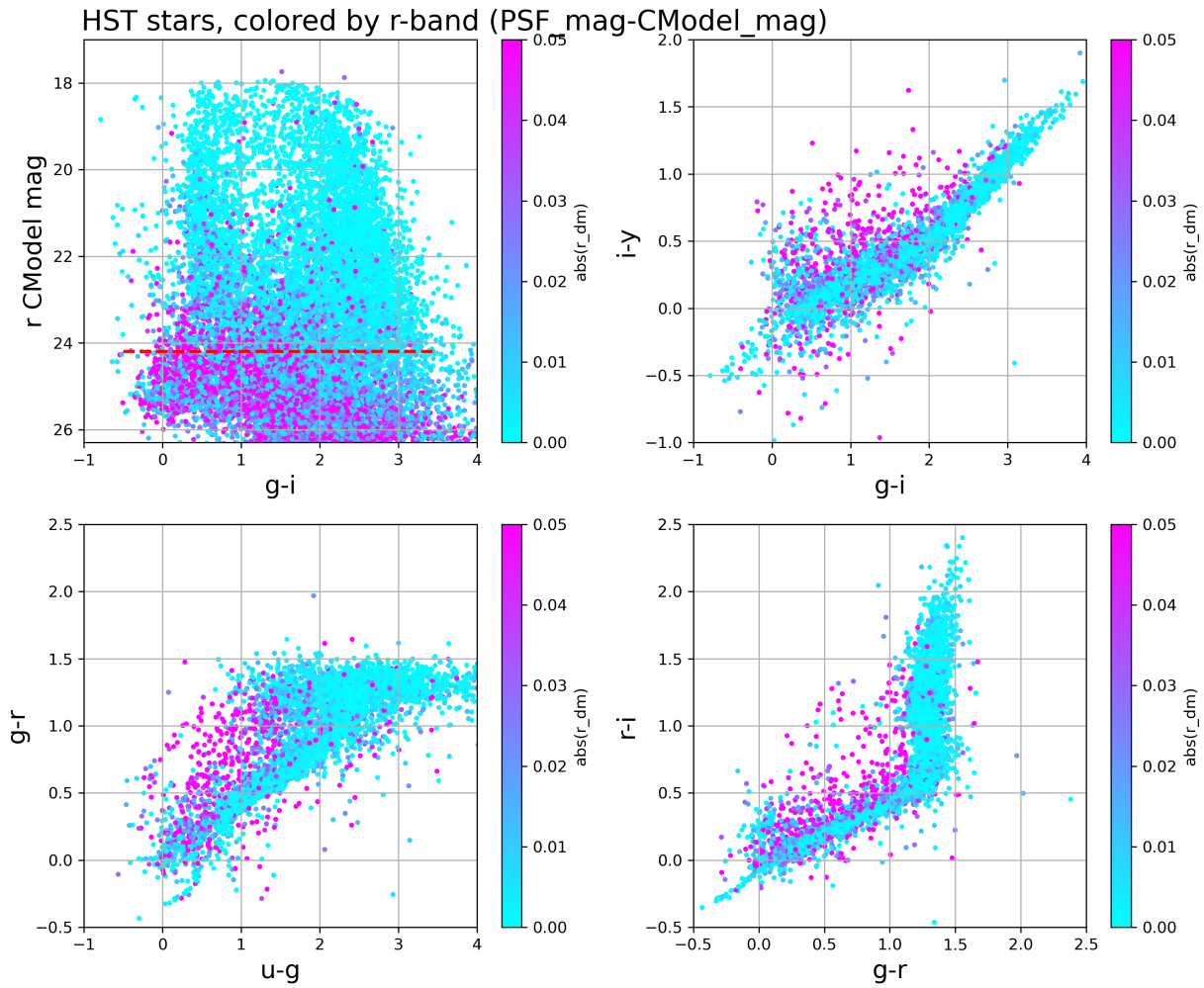


Figure 6. The distribution of objects classified as “stars” by HST, but which appear clearly resolved when using Rubin’s default `extendedness` parameter, in color-magnitude (top left) and informative color-color diagrams. Only objects with $r < 24.2$ are shown in the three color-color diagrams. Symbols are color-coded by Rubin’s Δm in the r band. The color-color distribution of objects with $\Delta m > 0.05$ mag (about 3% of all objects classified by HST as “stars” and brighter than $r = 23$) is inconsistent with main stellar locus and similar to that expected for barely resolved or unresolved binary stars.

A toy model for D distributions

Here we describe toy model... Figures 8 and 9

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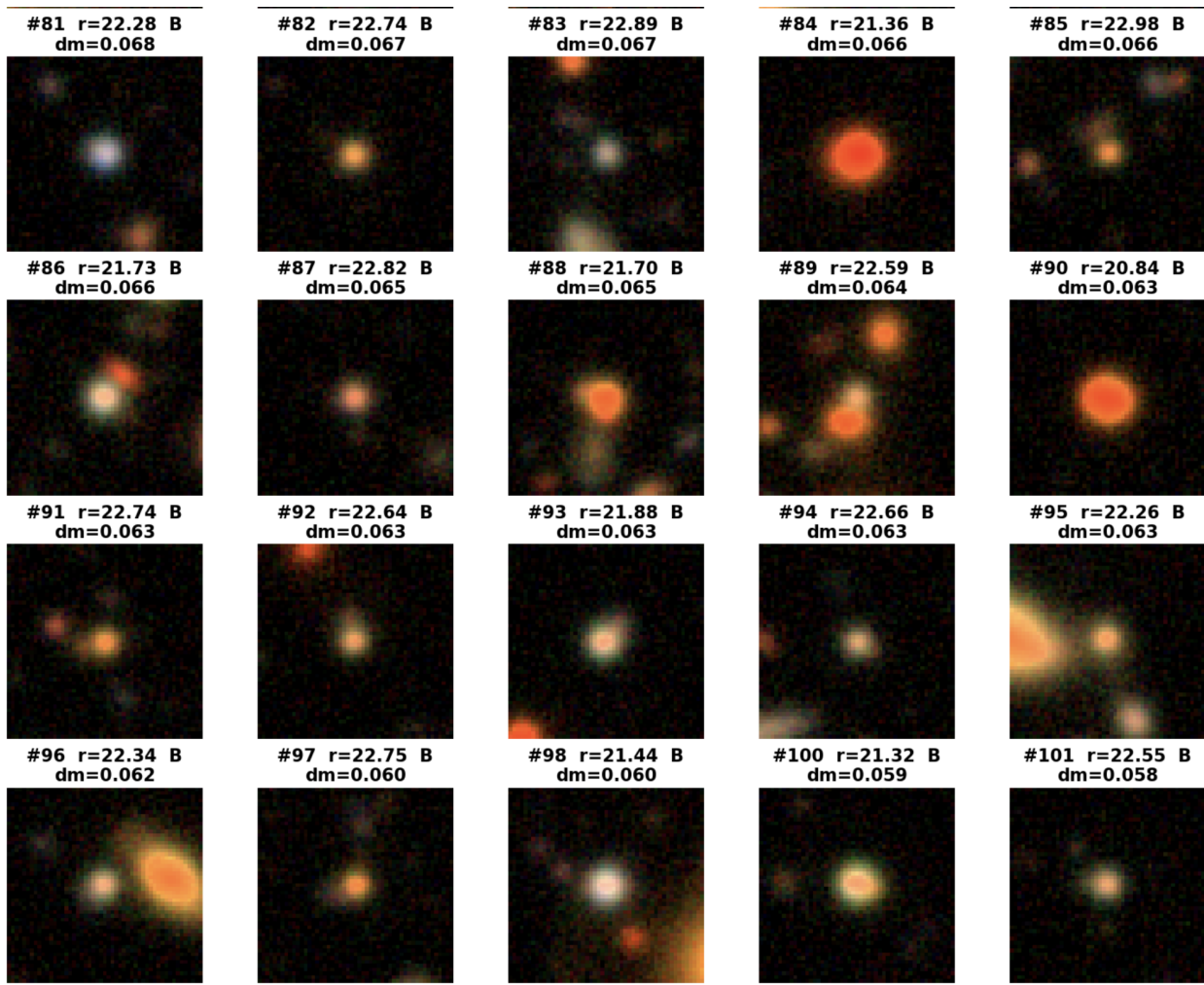


Figure 7. The Rubin *gri* color composite images extracted from “deep_coadd”, 10 arcsec by 10 arcsec (50x50 pixels), for a random subsample of 443 objects selected as “stars” in HST and resolved in Rubin, and brighter than $r = 23$.

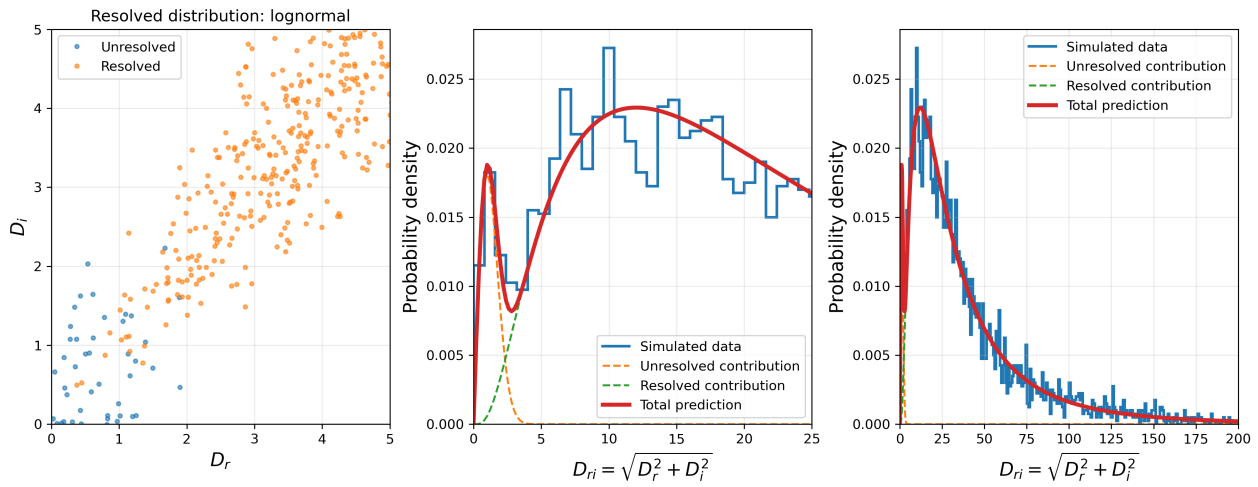


Figure 8. First figure...

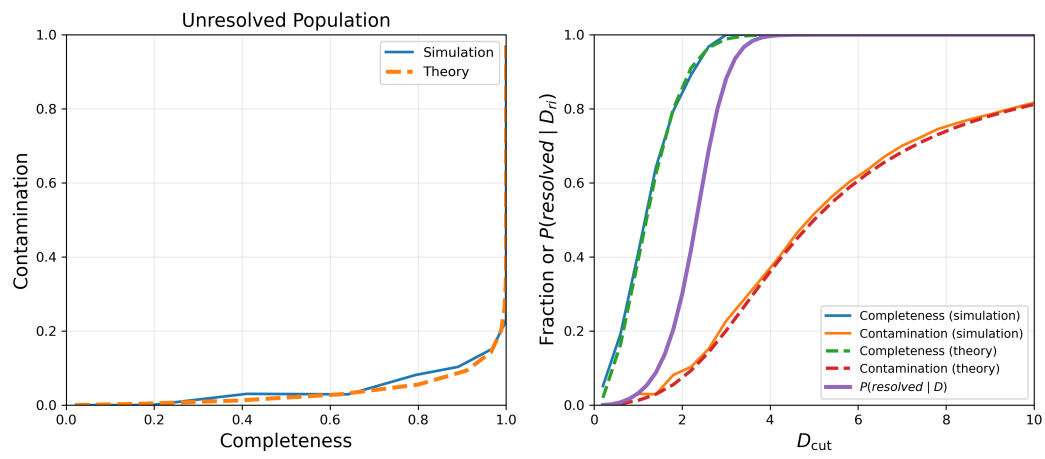


Figure 9. Second figure...

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